

Two Layers of an Economy: Decisions, Scripts, and the Anatomy of a Simulated Grocery Store

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Abstract

This paper tells the story of a simulated small grocery store from the ground up: not by walking through the software that builds it, but by walking through the economic decisions that give it its shape. Every number this construction produces — a receipt, an invoice, a monthly ledger, a tax filing — is the residue of somebody choosing something. A customer chooses between two cartons of milk. An owner chooses where to open a store before a single customer has walked through the door. The same owner chooses, later, whether to hire, and whether to expand, under capital he does not have room to lose. This paper follows each of those choices from the plain human intuition behind it to the precise mathematical object that intuition becomes, and does the same for the one thing in this world that is never a choice at all: the weather, the calendar, and the macroeconomic shocks that are scripted in advance and never listen to anything the simulated people do. That distinction — between what is chosen and what is merely scripted — is not a modeling convenience. It is what allows a later reader to ask what would have happened without the storm, or without the competitor, and receive an answer that is exactly true rather than merely estimated. The result is intended as a self-contained companion for anyone who wants to understand this simulated economy in its own right: every mechanism described here is described completely, in the same narrative register throughout, rather than deferred to outside reference material.

1 Two Layers of an Economy

Suppose you hand a new analyst a spreadsheet and tell her it is real. She has no way to check whether her conclusions are correct, only whether they sound reasonable, because nobody — not her, not you, not the business itself — actually knows the true mechanism that produced the numbers. Now suppose instead you hand her a spreadsheet you generated yourself, by drawing every column from a convenient distribution. She can check her work against your generating code, but there is nothing underneath the numbers worth discovering: no owner who misread a signal, no competitor who actually mattered less than everyone assumed, no honest mess in the paperwork to clean up before the real question can even be asked. Teaching data analysis with either kind of data teaches half the job.

This paper is about a third option: build the business first, as a world with people, budgets, and a street, and only afterward let the paperwork fall out of what that world actually did. Do that carefully enough, and you get something rare — a dataset messy enough to feel real, and yet one where every number still has a traceable cause, so a conclusion can be checked not just for plausibility but for truth.

1.1 An economy is a collection of decisions

Every column in a business's ledger is the fossil record of somebody choosing something: a customer choosing a shelf, an owner choosing a price, a bank choosing a loan. *If data is going to teach real analysis, it has to model the choosing, not the ledger.* This construction takes that literally. It contains no column that was not produced by a decision — a customer's, an owner's, or in one case a whole population's slow drift in and out of a neighborhood — and every decision is built the same way, in three steps that recur throughout this paper: first the plain economic intuition behind the choice, in language you would use to explain it to a friend; then the nature of the choice itself, meaning what kind of decision problem it structurally is, a discrete pick among a handful of options, a one-shot allocation of scarce capital, a threshold crossed or not; and finally the precise stochastic process that intuition becomes once it is written in the language of mathematics.

Meet Henrik Fjeld. He is not a real person — his name, his age, and the street he lives on are invented for each run of this simulation, chosen fresh from a small cast of possibilities every time someone generates a new store. But everything that happens to his shop is not invented at all: it is the exact, computed consequence of decisions made by simulated people responding to a simulated world, the same way a real ledger is the consequence of decisions made by real people responding to a real one. Consider one such decision, small enough to hold in your head. A customer walks into Henrik's shop wanting a carton of milk. Two sit on the shelf: a familiar national brand at €1.89, and the store's own cheaper label at €1.49. She does not compute anything — she just knows, the way you know at the supermarket, that today the cheaper one

will do just fine, but last week, buying for a dinner guest, only the brand name felt right. That is the entire economic content of the decision: a comparison between an option's price and how much she happens to want the thing the price difference buys. Section 3 turns that comparison into the single equation that governs every purchase in this economy, and Table 1 there shows the same customer, the same two cartons, and the one thing that actually changed between two different Tuesdays.

1.2 What is scripted and what is chosen must never be confused

Multiply Henrik's customer's decision by three hundred households, three years, and every shelf in the store, and you get something that looks, from the outside, exactly like a real small business: a ledger with good months and bad ones, a competitor who opens down the street and costs the owner something — though maybe not what he thinks it costs him — and a set of records honest enough to contain the ordinary mistakes a real bookkeeper makes. From the inside, though, every one of those numbers still traces back to a decision like Henrik's customer and her milk: a person, a comparison, a choice.

This is also where the model draws a line that is never allowed to blur. A storm, a tax change, a competitor's arrival are written into the world in advance, on a specific date, independent of anything anyone in the simulation does. A customer's purchase, an owner's price, an owner's decision to hire are never written down anywhere at all — they emerge, fresh, from the decision-maker's own situation each time the world is run. *What is scripted and what is chosen are never allowed to be confused.* Keeping that line sharp is what lets a later section ask “what would have happened without the storm?” and get back an answer that is exactly true, rather than merely estimated, because nothing about the storm could possibly have been caused by anything the shop's own people did.

Two layers follow from this one distinction, and this paper is organized around exactly the two of them. Section 3, **the micro layer**, walks through every kind of decision this simulated world contains: what a customer wants, what an owner believes before he has a single sale to go on, when an owner dares to expand, and who stays a customer and who drifts away. Nothing in this layer is written down in advance; everything in it is computed fresh, each time the world is run, from the situation the decision-maker actually faces. Section 4, **the macro layer**, covers everything that *is* written down in advance: the calendar, the weather, macroeconomic shocks, and the tax code the whole economy operates under. Section 5 explains why keeping these two layers from ever talking to each other in the wrong direction is what makes this world useful for more than description — it is what lets a later reader ask a genuine causal question and trust the answer. Section 6 covers a third thing that is neither a decision nor a script: the paperwork itself, which is allowed to be wrong in small, realistic ways even though the world underneath it never is. Section 2 places this construction against the other ways synthetic and case-study business

data are made today, and Section 8 steps back to ask what the whole architecture buys, and for whom.

The goal of everything that follows is not to prove a theorem. It is to leave a reader able to open Henrik Fjeld’s ledger, or the ledger of the next store this same machinery generates under a different roll of the dice, and already know, before running a single regression, what kind of story could possibly be hiding inside it.

2 Related Approaches and the Case for Structural Synthesis

Synthetic business data is not a new idea, and it is worth being precise about where this construction sits relative to approaches already in routine use. Four traditions dominate the current landscape, each built to optimize for something other than what this project optimizes for.

2.1 Rule-based generators

The simplest and most widely used tools, such as Faker and Mockaroo, fill a schema with plausible-looking field values: names, addresses, dates, amounts drawn from simple parametric distributions attached independently to each column. These tools are fast and entirely adequate for their actual purpose, populating a database or testing a software pipeline against realistically shaped inputs. They are not built to support analysis practice, because each column is generated independently of every other: there is no mechanism linking a customer’s income to their basket size, no mechanism linking weather to foot traffic, and no sense in which the resulting numbers were caused by anything. A student who regresses one generated column on another is not discovering a relationship, because none was ever built in.

2.2 Statistical and generative model synthesizers

A more sophisticated family of tools learns to imitate the statistical shape of a real dataset rather than sampling columns independently. Conditional generative adversarial networks such as CTGAN, variational autoencoders, and copula-based tools bundled in synthesis libraries such as the Synthetic Data Vault (Patki, Wedge, and Veeramachaneni, 2016; Xu, Skoularidou, Cuesta-Infante, and Veeramachaneni, 2019) are trained on a real seed dataset and learn to reproduce its marginal distributions and pairwise correlations closely enough to substitute for it in downstream use, often motivated by a differential privacy guarantee (Dwork, 2006). This is a genuine and valuable technology for privacy-preserving data sharing, and a poor fit for teaching data analysis, for three reasons. It requires an existing real dataset to imitate, so it cannot manufacture a genuinely new case. The generator itself is a black box, reproducing a distribution’s shape without asserting any mechanism that produced it, so there is no known ground truth against which a causal claim can be graded. And a distribution-matching model has no way to answer a

counterfactual question at all, because it was never given a reason for any number, only a shape to reproduce; asking such a model what revenue would have been without a tax change is not merely hard, it is category confusion.

2.3 The business case method

Business schools have long taught with hand-authored cases: a narrative description of a company's situation accompanied by a set of exhibits, built by a case writer around a specific pedagogical point. This tradition, associated most closely with Harvard Business School's case method, remains the gold standard for teaching judgment under ambiguity, and the persona-and-letter device this project's own case briefs use (Section 3) borrows directly from it. Its limitation is not narrative quality but process: each case is bespoke, manually tuned by its author to make the numbers come out right, produced once, and reused unchanged for years. There is no underlying generative process to interrogate, no way to produce a counterfactual twin of the same company under a different tax regime, and no way to grade a causal claim against a documented mechanism, because the case writer's spreadsheet is the only ground truth that exists, and it was authored to support a conclusion rather than to allow one to be discovered.

2.4 Structural microsimulation

The methodological tradition closest to this project's own is structural microsimulation, founded by [Orcutt's \(1957\)](#) proposal to model an economy as a population of individual units following explicit behavioral rules rather than aggregate equations, and carried forward today in tax and benefit microsimulation models such as EUROMOD and in agent-based market and transport simulations across applied economics. These models share this project's central commitment: agents obey genuine, documented behavioral rules, so a policy change can be replayed through an unchanged population and the difference is a true causal effect. Where they typically diverge is in what they are built to produce — an aggregated statistic destined for a policy briefing rather than a transaction-level business record — and because their audience is professional economists rather than students learning to reconcile data, they rarely simulate imperfect paperwork, since a policy model has no reason to misfile an invoice.

2.5 What this construction combines

No single tradition above combines all four of the properties this project requires at once: a fully documented structural generating process rather than a black box; transaction-level business artifacts carrying deliberately realistic, separately documented paperwork imperfections; exact counterfactual replay at the level of individual decisions, made possible only by the keyed-randomness discipline of Section 5; and a built-in scorecard for the analysis itself, three profit

figures (Section 3) that turn the abstract idea of analytical value into a specific, computed number — the gap between what an owner believed, what he actually earned, and what a perfectly informed operator could have earned. None of the four traditions above stage a protagonist who is wrong in a way that can be measured.

2.6 Value for teaching scientific reasoning, not only data analysis

The deeper claim behind this construction is that its principal teaching value is not technical fluency with a particular software tool but a disciplined scientific habit of mind, and that a dataset with a fully known generating process is uniquely suited to teaching that habit because it can tell a learner, definitively, when they are right. An analyst working this data is repeatedly placed in a position where a plausible-looking pattern is a trap rather than a finding: a flat category (Section 4) whose apparent seasonality is a budget-mediation effect rather than a genuine preference, a markdown campaign (Section 3) whose naïve before-and-after comparison is contaminated by the very selection rule that triggered it, an energy crisis (Section 4) whose use as a price instrument is subtly invalid because it also moves household budgets. Each trap has a documented, correct resolution, which means an instructor can teach the general scientific skill of interrogating one's own inference by handing a learner a specific claim, a specific method for checking it, and a specific ground truth that reveals whether the checking actually worked.

3 The Micro Layer: Economic Decisions from the Ground Up

Everything in this section shares one property: nothing in it is written down anywhere in advance. A customer's purchase, an owner's price, an owner's decision to hire, a household's decision to move away — these are computed fresh, each time the world is run, from the situation the decision-maker actually faces at that moment. This section walks through five such decisions, in the order a real small business actually encounters them: what a customer wants at the shelf; what an owner believes and decides before a single customer exists; how that same owner adjusts, day to day, once real customers do exist; when the owner dares to risk capital on growing the business; and, last, a kind of motion that is not really a decision at all but a population's slow drift, which this section includes anyway because it is exactly as emergent, and exactly as stochastic, as everything else here.

3.1 The customer's purchase decision

Consider Henrik Fjeld's customer again. She does not walk in with a spreadsheet. She walks in wanting milk, glances at two cartons, and buys one, and if you asked her why, she would

give you a sentence, not an equation: *the cheap one is fine today*, or *I want the good one, since I have a guest*. Every real purchase decision this store will ever record is, underneath, one of these small sentences, repeated by three hundred different people with three hundred different budgets and tastes, at every shelf in the store, every day for three years.

Structurally, this is what economists call a discrete choice: a customer facing a small, enumerable set of alternatives — the products on one shelf, plus the option of buying nothing here at all — picks exactly one. It is not a continuous decision like choosing how much to spend; it is a pick. What makes the pick interesting is that it depends on three separate things at once: how much the customer currently needs the category at all, how well each product on the shelf actually matches her taste, and how much each one costs relative to what she is willing to pay. A rich model of this decision has to hold all three considerations in the same comparison, rather than resolving need, taste, and price one after another as though they did not interact.

Put the sentence into the language of mathematics, and it becomes a single expression. Each product s offers a customer i a utility

$$U_{is} = \theta_{ic} + a_s + \gamma (1 - |b_i - \text{BrandLevel}_s|) - \beta_i \text{Price}_s,$$

where θ_{ic} is the customer's current need for the category. The term b_i is her brand affinity, BrandLevel_s is the product's own position on that same spectrum, and the middle term rewards a good match between the two, with γ setting how much that match matters. β_i is her own price sensitivity, and $a_s \sim \text{Normal}(0, 0.8)$ is an unmodeled per-product appeal intercept, standing in for the thousand unmodeled reasons one product outsells its shelf neighbor, such as packaging or familiarity. Alongside every stocked product sits an outside option, representing the choice to buy nothing here or to buy elsewhere, and the actual choice is drawn as

$$\text{choice} \sim \text{Categorical}(\text{softmax}(U_{is_1}, \dots, U_{is_K}, U_0)).$$

The softmax form is not a convenient approximation; it is the exact consequence of a well-known result in discrete choice theory. If each product's utility is disturbed by an independent, identically distributed Gumbel shock and the customer simply picks the option with the highest noisy utility, the resulting choice probabilities are exactly the softmax of the mean utilities — McFadden's random utility result (McFadden, 1974). The smooth formula above and the human story of a shopper with genuine but noisy tastes are the same model described two ways, and it happens to be exactly the model a real discrete-choice analyst would fit to purchase histories, so grading a later analytical exercise against the true parameters is a fair comparison of like against like.

Table 1 makes the same point in a different medium. It scores Henrik's customer's utility for both cartons on two different occasions — an ordinary Tuesday, and the evening she is shopping for a dinner guest — decomposed into exactly the terms the equation named.

	Ordinary Tuesday	Dinner guest
Brand match, $\gamma(1 - b_i - \text{BrandLevel}_{\text{brand}})$	0.60	0.60
Brand match, $\gamma(1 - b_i - \text{BrandLevel}_{\text{store}})$	0.50	0.50
Price term, $\beta_i \cdot \text{price difference}$	-0.40	-0.40
a_s (that day's unmodeled pull)	-0.05	+0.35
$U_{\text{brand}} - U_{\text{store}}$	-0.35	+0.45
Chosen carton	store brand	national brand

Table 1: *One customer, one choice, two occasions. Quality and price enter identically both times; only the unmodeled pull term changes — and it alone flips which carton wins.*

Nothing in the table is new information; it is the same equation, the same customer, and the same two cartons of milk, rearranged so that the one thing that actually changed — not her taste, not the prices, only the part of that day even she could not fully explain — is the one thing the eye lands on first.

This single structure produces, without any additional machinery, four properties a realistic market must have. Demand slopes downward, because raising a product's price lowers its utility and therefore its choice probability, at a rate set by each customer's own price sensitivity. Substitution follows a sensible pattern, because a price increase on one product pushes probability mass onto its nearest neighbors in brand level, not onto arbitrary alternatives. Budget discipline emerges naturally, because a customer works through a shopping list item by item until the budget binds. And hidden demand is recorded automatically: when a customer's first choice is out of stock, the original choice is logged as a latent truth before she re-evaluates the softmax over what remains, giving an honest ledger of demand that never appears in the sales data at all.

That last property deserves its own sentence, because it is what turns this decision from a single equation into a genuine daily market. A visiting customer's need for a category is driven by a pantry: a latent stock that drains at a steady rate and refills only when she actually buys more, $P_{ic,t+1} \stackrel{\leftarrow}{=} \max(0, P_{ict} - r_{ict}) + \text{units purchased}$, with the drain rate itself scaled by the season's demand modifier rather than added to her desire directly, so a hot week does not make a household want ice cream in the abstract, it makes the household eat the ice cream in the freezer faster. Her need for the category, $\theta_{ict} = \alpha \max(0, P_{ic}^{\text{tgt}} - P_{ict}) / P_{ic}^{\text{tgt}}$, is exactly the shortfall against a target of ten days' comfortable cover, and it is this shortfall, not a fresh random draw, that feeds θ_{ic} in the utility equation above. When her chosen product is unavailable, three things happen in order: the original intent is logged as a latent truth under the cause `stockout`; the same softmax is re-evaluated with that product removed, so probability mass flows to its nearest neighbors exactly as the substitution story above predicts; and if the re-evaluated choice still lands on the outside option, the store has lost a sale it will never observe in its own data. Four distinct causes of invisible demand accumulate this way over a year — an arrival outside opening hours, a stockout, an exhausted weekly budget partway through the list, and the outside

option simply winning on price — and together they are the honest, mechanistic version of the familiar claim that sales data are demand data, censored in several distinct ways, each with its own recorded cause.

3.2 The owner’s opening decision

Before Henrik Fjeld’s shop can sell a single carton of milk, Henrik himself has to make a decision unlike any his customers will ever face: where to open, what to stock, how much of it to buy, and how many people to hire, all at once, with a fixed sum of savings and no sales history whatsoever to consult. A customer’s decision is repeated thousands of times and can afford to be a little wrong on any given Tuesday; an owner’s opening decision happens exactly once, commits capital he cannot easily get back, and has to be made entirely from belief rather than observation.

This is not a discrete pick among a handful of alternatives, and it is not a repeated decision that averages its own mistakes away over time. It is a one-shot, capital-constrained allocation problem: several interdependent choices — which location, which products, how much of each, how many staff — have to be made together, because each one tightens or loosens the constraints on all the others, and all of them draw on the same finite pool of capital. The single feature that most separates this decision from the customer’s is that it is optimized against a *belief*, not against the truth, because the true demand a location could support is a fact about the world that nobody, least of all the owner standing in an empty storefront, can yet observe.

The true demand potential of a location for a merchandise category is $D_{cl} = \text{Households}_l \cdot \mu_c$, where μ_c is the mean monthly per-household consumption a store of this kind could realistically capture. This is ground truth; nobody in the model observes it directly. Henrik’s belief is the truth seen through a noisy and mildly optimistic lens, $\hat{D}_{cl} = D_{cl} e^{\delta_c}$ with $\delta_c \sim \text{Normal}(0.10, 0.25)$. The error lives on the logarithmic scale because people misjudge quantities in ratios rather than in absolute units — nobody thinks their estimate is off by thirty-seven boxes, they think demand is probably a fifth higher or lower than it looks — and the positive mean encodes a documented behavioral regularity: new business owners tend to be mildly overoptimistic. This single choice is what causes the store to systematically overstock in its early months, and the size of that overstocking can be computed exactly from the belief error rather than asserted as an unexplained fact about the data.

With beliefs, costs, and a handful of standing policies fixed — a fixed markup m_c per category, a flat safety-stock rule holding a fraction $\eta = 0.30$ of a month’s believed demand on the shelf at all times, and a rule that no single product should capture more than a share $\rho = 0.05$ of its category’s demand — Henrik’s opening decision is a mixed integer linear program, choosing simultaneously which location to occupy, which products to list, how much of each to buy, and how many staff to hire, so as to maximize the profit he believes he will earn:

$$\max \sum_c \sum_{s \in S_c} (1 + m_c) \text{Cost}_s \cdot u_s - \text{TotalCost}_0$$

subject to

$$\begin{aligned} \sum_l y_l &= 1 && \text{(exactly one location)} \\ u_s &\leq q_s && \text{(cannot sell what is not stocked)} \\ \sum_{s \in S_c} u_s &\leq \sum_l y_l \hat{D}_{cl} && \text{(believed category demand)} \\ u_s, q_s &\leq \rho \sum_l y_l \hat{D}_{cl} && \text{(no product dominates its category)} \\ \sum_{s \in S_c} q_s &\geq \eta \sum_l y_l \hat{D}_{cl} && \text{(the safety-stock rule)} \\ \sum_s q_s &\leq \sum_l y_l \text{ShelfCapacity}_l && \text{(shelf capacity)} \\ \sum_s x_s &\leq \sum_l y_l \text{ShelfSlots}_l && \text{(shelf slots)} \\ E_0^h &\geq \sum_l y_l \text{OperationalNeeds}_l - 1 && \text{(minimum staffing)} \\ \text{TotalCost}_0 &\leq \text{InitialCapital}_0 && \text{(capital)} \end{aligned}$$

with $y_l \in \{0, 1\}$ marking the chosen location, $x_s \in \{0, 1\}$ marking which products are listed, $q_s \geq 0$ the quantity purchased, $u_s \geq 0$ the quantity Henrik believes he will sell, and $E_0^h \geq 0$ the additional staff hired beyond himself. A mixed integer program is the right tool here, rather than a simpler heuristic, precisely because the decision genuinely is joint: location and listing choices are binary, purchase quantities are continuous, and every constraint linking them is linear once the location is fixed. One detail matters more than its size suggests: Henrik draws no wage in this accounting, since a sole proprietor is paid out of profit, not payroll, and a numerical audit confirmed that charging him a market wage for his own long hours makes every affordable opening plan show a believed loss, which no rational entrepreneur would actually sign.

This one optimization produces several properties without any of them being separately engineered. Location choice becomes a genuine trade-off between rent and believed demand. Product variety emerges from the interaction of beliefs, capital, and the no-domination constraint, rather than being a hand-set number. And because the belief error above carries a positive mean, the optimal solution systematically overstocks relative to the truth, so that when an analyst later finds bloated inventory in the cost sheets, there is a genuine mechanism behind it rather than an injected anomaly. Figure 1 lays out the whole causal chain: everything upstream of the decision

node is generative ground truth Henrik never sees directly; the decision node and everything downstream of it is his boundedly rational response to a distorted and simplified picture of that truth.

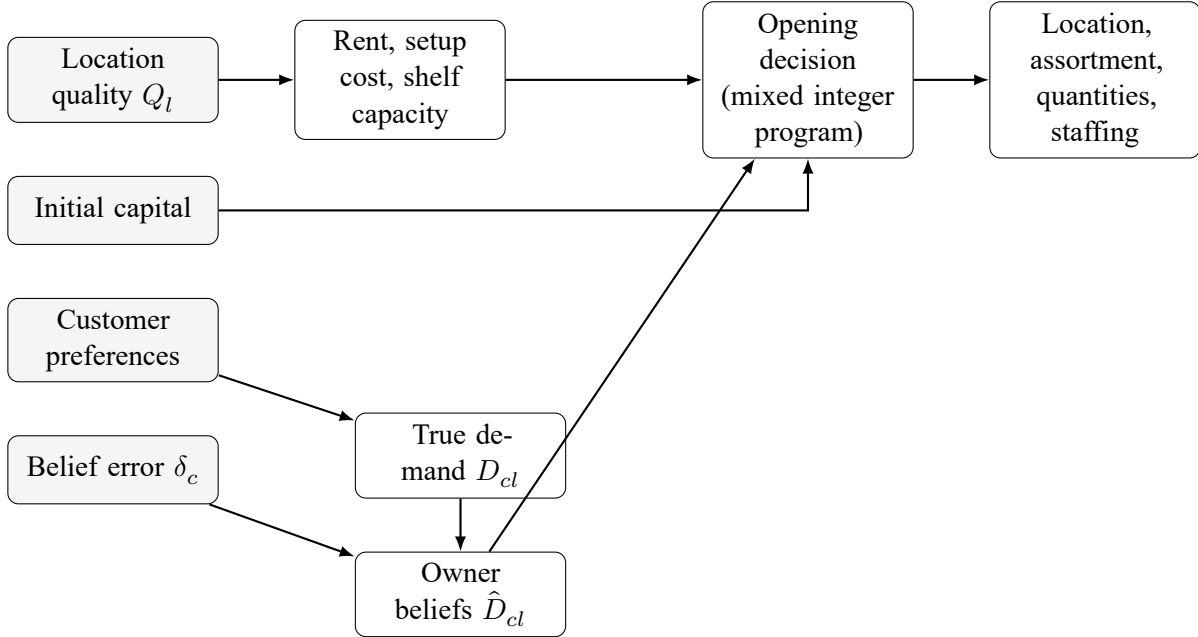


Figure 1: *The causal skeleton of the opening decision. Everything upstream of the decision node is generative ground truth; the decision node and its consequences are the owner’s boundedly rational response to it.*

Five specific divergences between belief and truth are built into this one decision, and each is a documented, discoverable fact rather than an arbitrary flaw: optimistic demand beliefs, a flat safety-stock rule applied regardless of season, category-level markups that ignore individual price sensitivity, static beliefs that never anticipate the seasonal cycle of Section 4, and a flat product-share cap that ignores genuinely heterogeneous tastes. Section 3.3 shows what happens once Henrik starts learning from real sales instead of belief alone, and Section 5 shows exactly how much that gap between belief and truth ends up costing him.

3.3 The owner’s day-to-day adaptive policies

Once the shop is open, Henrik stops guessing and starts watching. He cannot see true demand any more now than he could on opening day — he only ever sees what actually sold, which is not the same thing, since a stockout hides the sale that would have happened. What he does instead is what any real shopkeeper does: he keeps an eye on what has been selling lately, orders roughly that much again, marks down whatever is piling up on the shelf, and worries about cash before he worries about the annual accounts.

This is a genuinely different kind of decision from the opening allocation, even though the same person makes both. It is repeated weekly rather than made once, it responds to observed

history rather than to a belief formed in a vacuum, and it is deliberately a heuristic rather than a re-optimization: Henrik does not re-solve a mixed integer program every Monday morning, he applies a simple, realistic rule of thumb, the same way a real shopkeeper does, and the rule's own blind spots are as important to the model as its logic.

Once a week Henrik places a restocking order, sized off the simplest forecast a real shopkeeper actually runs: a trailing four-week average of his own recorded sales,

$$\hat{D}_{cw}^{\text{op}} \leftarrow \frac{1}{4} \sum_{j=1}^4 \text{Sales}_{c,w-j}.$$

He orders up to a target sized for the review and lead time ahead of him, plus his standing safety-stock rule, $\text{Target}_{sw} = \left(\frac{R+L}{7} + \eta \frac{30}{7}\right) \hat{D}_{cw}^{\text{op}} \tilde{\sigma}_{sw}$, with $R = 7$ days between reviews, $L = 2$ days of lead time, and $\tilde{\sigma}_{sw}$ the product's smoothed recent share of its category. This rule carries two realistic flaws built directly into its structure. It is backward looking, so a rising seasonal trend is always underestimated by the average of the weeks before it rose. And it is blind to censoring, because the sales it averages are themselves truncated by the store's own past stockouts, producing a self-reinforcing censoring spiral: a product that stocks out repeatedly reports weak sales, earns a smaller share of the next order, and stocks out again.

Henrik also runs two distinct kinds of promotion, and the contrast between them is itself an intended lesson. A markdown campaign triggers endogenously whenever a category's weeks of cover exceeds a threshold, targeting the slowest-selling third of that category. Because the trigger is overstock, and overstock correlates with weak underlying demand, any naive before-and-after read of a markdown's effect is contaminated by exactly the selection problem that afflicts real-world marketing analytics. A separate, calendar-triggered storewide discount on the last Sunday of every month is, by contrast, entirely exogenous — announced in advance and correlated with nothing about current stock — providing a clean source of price variation before anyone tries to untangle the confounded markdowns.

Perishable stock spoils overnight,

$$\text{Spoiled}_{st} \sim \text{Binomial}(\text{OnHand}_{st}, \min(0.35, \lambda_{c(s)}^d f_{c(s),t})),$$

with the spoilage factor f_{ct} built from the same log-additive logic as every other multiplier in this paper: a heat term strongest for ambient perishables such as bread and produce, a cold-chain strain term loading on the macro shocks of Section 4 and strongest for frozen and dairy goods, and an idiosyncratic batch-luck term standing in for the reality that one delivery simply travels worse than another. And Henrik's cash position is tracked explicitly, with a credit line available whenever a month's close would otherwise take cash below zero; restocking is itself capped by available cash and credit headroom, so a struggling store visibly thins its own shelves months before any trouble would show up in an annual profit figure, exactly as a real distressed retailer's inventory often signals difficulty well before its accounts do.

3.4 The owner's expansion decision

Sometime in Henrik's second year, if the business has been kind to him, he faces a decision every real small-business owner eventually faces: whether to risk some of what he has built on growing further. He does not commission a discounted-cash-flow analysis of the decision. He watches his own savings account. When it has grown past a point that feels safe to risk, and only then, he hires his first employee, extends his hours, and deepens his shelves.

This is neither the one-shot optimization of the opening decision nor the weekly heuristic of ordinary restocking. It is a threshold decision under genuine uncertainty about the future, and the threshold is framed around what Henrik can actually afford to lose rather than around a forecast of what the expansion will return — much closer to what entrepreneurship research calls *effectuation* (Sarasvathy, 2001) than to a textbook net-present-value calculation: a real small-business owner reasons from the resources already in hand and the loss he can tolerate, not from a spreadsheet of projected returns he has no reliable way to build.

From the beginning of the second year, Henrik retains a share $\rho_{RE} = 0.5$ of every profitable month's after-tax result as declared retained earnings, drawing the rest out as his own income, $\text{draw}_m = (1 - \rho_{RE}) \max(0, \pi_m)$, and neither draws nor retains anything in a loss month. Once retained earnings and available cash both cross fixed thresholds, $\text{RE}_m \geq K_{\text{exp}}$ and $\text{cash}_m - \text{tax_reserve}_m \geq C_{\text{exp}}$, he expands on the first day of the following month, paying a one-time fit-out cost from savings.

The threshold K_{exp} is calibrated so that, under the realized path of the model, the expansion lands in the autumn of the second year — after a scripted local population increase but before a scripted competitive entry the following year — producing a specific and analytically important narrative: the expansion looks entirely sensible at the moment it is made, based on everything Henrik could reasonably have known at the time, and only becomes questionable in hindsight once a later, unrelated shock arrives. A counterfactual twin of the model, in which the expansion threshold is disabled entirely, reveals that the expansion's true cost across its first fourteen months substantially exceeds the additional margin the extra hours and capacity actually generate, because the fixed cost of an additional wage is paid in full while the additional revenue it produces is diluted by the ordinary grocery margin. This is a deliberately reproduced version of one of the most common capital-allocation mistakes in small retail — investing at what retrospectively turns out to be the peak of the opportunity — generated here from the interaction of an ordinary retention rule, an ordinary calibrated threshold, and an otherwise unrelated later shock, rather than scripted as an explicit narrative choice.

3.5 Population dynamics: the customer panel as a stochastic flow

Not every micro-level movement in this world is a decision in the sense the last four sections meant. A neighborhood is not static: households move in, households move out, and a shop's

regulars are a population in slow motion, not a fixed cast of characters. This is included in the micro layer anyway, because it is exactly as emergent and exactly as stochastic as everything above it, even though nobody is really “choosing” to leave the way Henrik chooses a price.

Structurally this is a renewal process: each customer has a type fixed at creation that determines whether they are subject to departure risk at all, departures happen at a constant hazard rate rather than on a fixed schedule, and each departure is eventually replaced, so the population’s size is roughly stable even as its individual membership turns over completely.

Each regular customer is given a persistence type drawn once, rooted with probability 0.80 and transient with probability 0.20. A rooted customer never leaves the neighborhood within the three-year window at all; a transient customer faces an ongoing monthly hazard of departure $h = 1/18$, a mean tenure of eighteen months. Every departure is matched, after a delay of $1 + \text{Geometric}(0.5)$ months, by a fresh household moving into the vacated home, and a slow trickle of genuinely new arrivals, $\text{Poisson}(4)$ per year, represents modest organic growth on top of straightforward replacement. Newly arrived customers are drawn with an even fifty percent chance of either persistence type, reflecting the reasonable assumption that a household which has just moved once is more likely to move again than the settled population it replaces.

Nothing about this churn is announced anywhere in the visible data; a customer’s card token simply stops appearing in the receipts one month and never returns, and a new token begins appearing elsewhere. An analyst must infer departure from silence, taking care not to confuse a customer who has genuinely left with one merely visiting less often for other reasons — a distinction that becomes especially difficult near the very end of the observed window, where a recently silent customer cannot yet be told apart from one who will return next month. This is a form of the right-censoring problem familiar from survival analysis, and it is a genuine, unavoidable feature of any panel observed for a finite length of time, reproduced here rather than glossed over.

4 The Macro Layer: The Deterministic Script

Everything in the previous section was computed fresh, each time the world is run, from a decision-maker’s own situation. Everything in this section is the opposite: written down in advance, on a specific date, completely independent of anything any customer or owner in the simulation does. Weather does not care whether Henrik had a good month. A competitor does not time its entry to Henrik’s cash position. This independence is not a simplifying convenience; it is what allows weather to function as a natural experiment and a cost shock to function as an instrument in later analysis, because nothing about the shop’s own behavior could possibly have caused the weather.

4.1 Calendar and weather

The simulated year runs three hundred sixty-five days, aligned to a real calendar so seasons, holidays, and an analyst's ordinary intuitions all line up, including two days on which the store is legally closed. Temperature follows a deterministic annual cycle plus a persistent stochastic anomaly, $T_t = \mu_T(t) + \varepsilon_t$ with $\mu_T(t) = 12 + 9 \cos\left(\frac{2\pi(t-200)}{365}\right)$ and $\varepsilon_t = \phi \varepsilon_{t-1} + \text{Normal}(0, \sigma_\varepsilon)$, $\phi = 0.7$, an autoregressive form chosen because it is the maximum entropy stationary process consistent with exactly the two facts about weather worth encoding: a stable typical spread, and warm or cold spells that persist for several days rather than resetting each morning. The decomposition into a deterministic cycle and a stochastic anomaly is itself a causal design choice: it lets a later analysis distinguish the predictable effect of season from the genuinely surprising effect of weather, since the anomaly is built orthogonal to the calendar, while in raw data the two are thoroughly entangled, because July is warm precisely because it is July.

Rainfall needs two separate decisions, whether it rains at all and how much, each with its own natural distribution. Occurrence follows a two-state Markov chain, $W_t \mid W_{t-1} \sim \text{Bernoulli}(p_{01}(1 - W_{t-1}) + p_{11} W_{t-1})$ with $p_{01} = 0.25$, $p_{11} = 0.55$, because weather fronts pass through and wet days genuinely cluster, which an independent coin flip at every day cannot represent; these transitions imply a stationary wet-day frequency of about thirty-six percent and wet spells averaging just over two days, both ordinary for a temperate climate. Given rain, the amount follows $R_t \mid W_t=1 \sim \text{Gamma}(1.4, \text{mean} = 5 \text{ mm})$, the standard meteorological choice, producing the realistic pattern of many light drizzles and few genuine downpours.

4.2 Category demand and foot traffic

A daily, per-category demand modifier M_{ct} scales how strongly the neighborhood wants a given category on a given day, built log-additively from three genuinely distinct causal channels, $\log M_{ct} = a_c \cos\left(\frac{2\pi(t-200)}{365}\right) + \kappa_c z_t + h_c H_t$, where z_t is the standardized temperature anomaly and H_t flags days before a major holiday. The seasonal amplitude a_c captures the predictable annual cycle; the anomaly loading κ_c captures the pure weather effect, an unseasonably warm week moving demand right now regardless of the calendar; the holiday loading h_c captures the stock-up surge before a major holiday. One category is deliberately given all-zero loadings, so nothing in preferences makes it more or less wanted at any time of year; its observed sales are nevertheless not perfectly flat, because every category draws from the same household budget, and when other categories surge seasonally they crowd the flat category out. An analyst who finds an apparent seasonal pattern here and stops has mistaken a budget-mediation effect for a genuine preference effect. A small number of specific product types whose seasonality flagrantly contradicts their category's average, such as ice cream, receive their own finer-grained tilt, $\log \psi_{pt} = a_p \cos\left(\frac{2\pi(t-200)}{365}\right) + \kappa_p z_t$, entering the discrete-choice utility of Section 3.1 directly as $U_{ist} = U_{is} + \log \tilde{\psi}_{p(s),t}$.

Weather does not only change what people buy; it changes whether they come at all. A daily traffic modifier, $\log \Lambda_t = -0.20 W_t + 0.15 H_t + \varepsilon_t^\Lambda$ with the same autoregressive persistence as temperature ($\phi^\Lambda = 0.85$), scales every customer’s visit probability down by about eighteen percent on a wet day and up by about sixteen percent before a major holiday. A subtle consequence follows automatically from the pantry mechanism of Section 3.1: because a customer’s need for a category grows with time since their last purchase, a customer rained off on one day simply arrives hungrier a few days later. The familiar retail pattern of a dip on the rainy day followed by a rebound emerges purely from two mechanisms that never reference each other, with no separate rebound parameter anywhere in the model.

Household budgets move too. Each week’s budget is $B_{iw} = B_i \cdot e^{\xi_{iw}} \cdot s_{iw} \cdot g_{iw}$, with an ordinary weekly wobble $\xi_{iw} \sim \text{Normal}(0, 0.12)$ and two rarer, more consequential terms: a tight spell sets $s_{iw} = 0.65$ for a genuine hardship such as a job loss, beginning with probability $0.004(1 + 2\bar{g}_w)$ per week and persisting a memoryless Geometric(1/6) further weeks, where $\bar{g}_w$ is the scripted energy crisis’s own trajectory below; a splurge sets $g_{iw} = 1.5$ for a single week with probability 0.01. The tight spell’s entry rate is deliberately coupled to the crisis trajectory, tripling at its peak, because a real energy crisis squeezes household heating bills at the same time as it squeezes a retailer’s refrigeration costs.

4.3 Macroeconomic cost shocks

Beneath the everyday weather sits a slower-moving layer of wholesale cost shocks, and beneath that a continuous inflation drift, $\text{infl}(t) = e^{\pi t/365}$ with $\pi = 0.025$ a year. Shocks proper arrive as a marked point process: $K \sim \text{Poisson}(1.5)$ events, each at $\tau_k \sim \text{Uniform}(1, 365)$, a Poisson count because it is the maximum entropy distribution for a number of rare independent events given only a rate, with a well-known property of the Poisson process guaranteeing that, given the count, arrival times are conditionally uniform across the year — a theorem, not an additional assumption. One large event, an autumn energy crisis, is scripted rather than drawn at random, so every run tells a genuinely interesting supply-side story. Each event’s peak magnitude is $\zeta_k \sim \text{LogNormal}(\log \zeta_0, 0.3)$, and its effect over time is $g_k(t) = \zeta_k \cdot \min(1, (t - \tau_k)/14) \cdot \exp(-(t - \tau_k - 14)_+/\rho_k)$, ramping up over two weeks and decaying exponentially, neither a permanent level change nor a transient blip but a slow wave a forecaster genuinely has to track. Every affected category’s cost multiplier then assembles the drift and every live shock, $\log \text{CostMult}_{ct} = \pi t/365 + \sum_k g_k(t) \mathbb{1}[c \in A_k]$.

Henrik never observes CostMult_{ct} directly; he sees only his supplier’s invoices, and tracks an exponentially smoothed trend, $\overline{\text{Cost}}_{s,t} = \alpha \text{Cost}_{s,t} + (1 - \alpha) \overline{\text{Cost}}_{s,t-}$, $\alpha = 0.35$, moving the shelf tag only once the smoothed cost has drifted more than three percent from the current price — menu-cost pricing, a pattern well documented in the economics literature. The consequence is that shelf prices vary over time for reasons entirely unconnected to the demand side

of the market: a foreign harvest failure moves costs without touching any household's budget, exactly the exogenous price variation an econometrician needs to estimate a genuine price elasticity rather than a number contaminated by reverse causation. The scripted energy crisis is a deliberate exception to this cleanliness, since it also squeezes household budgets through the tight-spell mechanism above, making it a subtly invalid instrument whose bias can, in principle, be computed exactly.

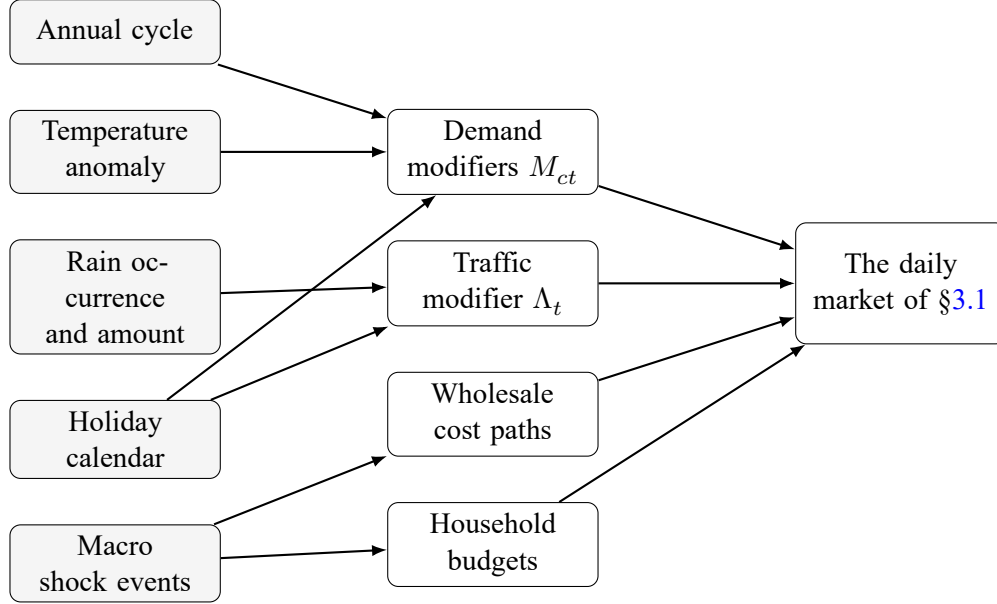


Figure 2: *The causal skeleton of the macro layer. Every arrow points into the daily market that Section 3.1 performs; nothing points back, which is what licenses treating these channels as exogenous instruments and natural experiments.*

4.4 Taxation as scripted policy

A policy laboratory without taxation would be unable to represent the most common lever through which real macroeconomic policy actually reaches individual households and businesses, so a complete, differentiated tax system sits in the baseline world itself, exactly as scripted and exactly as exogenous as the weather above. Value-added tax applies at a category-specific rate $r_c(t)$, and Henrik's shelf tag is formed from the tax-exclusive, smoothed invoice cost with the current rate reapplied on top, $\text{Tag}_{st} = \text{charm}(\overline{\text{NetCost}}_{s,t} \cdot (1 + m_c)(1 + r_c(t)))$. Because the rate enters multiplicatively on the trend he already tracks, a rate change reaches his *intended* price immediately and exactly, while the price customers actually see continues to lag through the same menu-cost mechanism above — exactly the kind of incomplete, delayed tax pass-through documented in the empirical public finance literature. Value-added tax is remitted monthly as the difference between output tax on sales and input tax on purchases, $\text{VAT}_t = \sum_c \text{Sales}_{ct} r_c / (1 + r_c) - \sum_c \text{Purchases}_{ct} r_c / (1 + r_c)$; payroll tax applies to wages at

a flat employer rate $r^{\text{pay}} = 0.25$; and a flat tax applies to positive annual profit only, accrued in the year it is earned, $\text{ProfitTax} = r^{\pi} \cdot \max(0, \text{ProfitBeforeTax})$, $r^{\pi} = 0.20$.

Henrik is tax-aware but not tax-strategic: he correctly prices on his net-of-tax margin and correctly accounts for the full employer cost of hiring in his opening decision, but he never performs cross-category tax arbitrage or elasticity-aware repricing in response to a tax change, leaving that gap as an open door for a later prescriptive analysis to identify and price. One piece of this is provably harmless rather than merely assumed away: because profit tax is a flat rate on a positive base, $\arg \max_{\text{decisions}} (1 - r^{\pi}) \pi(\text{decisions}) = \arg \max_{\text{decisions}} \pi(\text{decisions})$, so an owner who ignores profit tax when making ordinary operating decisions loses nothing relative to one who optimizes around it. No such guarantee holds for value-added tax, whose category-specific rate changes relative prices across categories and can, in principle, reward cross-category arbitrage Henrik never performs.

4.5 A world that stays unquiet

A scenario, in this model, is nothing more than a declarative edit to one of the quantities this section already defined: an extra cost event riding the same trajectory $g_k(t)$ above, a multiplier on the traffic modifier Λ_t , a multiplier on household budgets B_{iw} , or a new tax rate path $r_c(t)$ from a given date. A scenario never edits an underlying random-number stream directly, only the script. Consistent with the design principle that a three-year history should not read as a single stable year merely repeated three times, the second and third years of Henrik’s story are populated with a specific sequence of such edits, each chosen to test a distinct analytical skill: an equipment failure destroying frozen and refrigerated stock overnight, a single-category cost shock on dairy, a prolonged summer heatwave, a scripted local population increase, a contractual rent increase at a two-year lease renewal alongside similarly lumpy, dated steps in wages and utilities, and the entry of a discounting competitor a short distance away.

That competitor’s effect on individual customers is deliberately heterogeneous,

$$m_i = \exp(-\chi \cdot \tilde{s}_i \cdot k_i),$$

a multiplicative penalty on visit probability where $\tilde{s}_i$ is a customer’s percentile rank in price sensitivity and $k_i = 1.0$ for transient customers but only 0.6 for rooted ones, so the competitor draws away price-sensitive customers fastest, and customers held by relationships rather than price alone are penalized at only sixty percent of the rate a transient customer at the same price sensitivity would be. It is followed by a scheduled, rather than reactive, pricing response, a short seasonal street festival, and a second cost shock on staple goods.

5 Why the Split Licenses Counterfactual Reasoning

Every decision in Section 3 and every script in Section 4 was built the way it was for one reason that this section finally makes explicit. Because the macro layer is written entirely in advance and never listens to anything the micro layer does, a counterfactual world can be produced almost for free: edit the script — delete the scripted energy crisis, disable the expansion threshold, remove the competitor entirely — and replay the identical daily market mechanism with every other random draw held fixed. The difference between the original run and the edited run is then the causal effect of the edit itself, not merely an estimate of it.

This only works because of a discipline that has been present, silently, in every equation in this paper: every random draw is keyed by the stable identity of the entity it concerns, a specific customer on a specific day, a specific product on a specific night, using independently spawned child streams, rather than one long sequential stream. If every draw were taken sequentially, deleting a single event partway through the year would shift every subsequent draw, and the comparison between two runs would be contaminated by reshuffled randomness that has nothing to do with the edit itself. Keying every draw by identity means a customer untouched by the edit reproduces the identical decision, bit for bit, in both runs — the common random numbers technique familiar from the simulation literature, adopted from the first line of the implementation rather than retrofitted later, because retrofitting it afterward would require rewriting every random draw in the model.

Two concrete pairs of counterfactual twins in Henrik’s own three-year story cash this discipline out into an actual number rather than a methodological promise. One twin disables the expansion threshold of Section 3.4 entirely; comparing it to the real run reveals that the expansion’s true cost across its first fourteen months substantially exceeds the additional margin it generates. A second twin removes the competitor of Section 4 entirely; comparing it to the real run prices exactly what the competitor cost Henrik, as opposed to what he believes it cost him. A specific, intended finding of the model is that the capital decision very often costs the business substantially more than the competitive entry does, even though the competitor is the more visible and more emotionally salient event to an owner living through it — which is exactly why an instructor can hand a learner Henrik’s own conviction that the competitor is to blame, and let the two counterfactual twins settle the question the data, rather than the instructor, gets to answer.

With a full year performed, three profit figures become computable side by side: what Henrik believed he would earn at the moment of opening, π_{believed} , read directly off the opening decision’s own objective value; what the ledger actually shows after a full year of realistic behavior, π_{realized} , the number a real accountant would certify; and what a perfectly informed operator could have earned given the same world — the same customers, the same script, the same policy shapes, but true demand in place of Henrik’s censored, backward-looking forecast — π_{oracle} .

The two gaps between these figures are exact differences between fully computed runs of the same mechanism, $\Delta_{\text{optimism}} = \pi_{\text{realized}} - \pi_{\text{believed}}$ and $\Delta_{\text{analytics}} = \pi_{\text{oracle}} - \pi_{\text{realized}}$, where the first measures the cost of Henrik’s initial overoptimism and the second measures the value of better information and better decisions — the single number this entire construction exists to make credible by actually computing it, rather than merely asserting a plausible size for it.

6 Recording and Accounting: An Imperfect Window on a Perfect World

Everything described so far — every decision, every script — happens inside a simulated reality that is always exactly conserved: every unit that is bought is eventually sold, spoiled, or counted, and every euro that moves is accounted for somewhere. An early, fully reconciled version of this dataset was reviewed and found, on inspection, to have one serious flaw: its records were perfect. Revenue tied to the cent, every inventory identity closed exactly, and no document was ever missing, duplicated, or mistyped. Real bookkeeping is never like this, and a dataset that reconciled perfectly would itself be evidence of fabrication to any experienced reader.

The paperwork therefore sits in a third layer of this model, neither a decision nor a script: a small number of realistic paperwork defects, injected at low, carefully calibrated rates only at the point of export, never touching the simulation’s own internal state. A point-of-sale terminal retries an upload and duplicates a till receipt. A distracted clerk posts the same invoice twice. A delivery is paid for but never logged. Unrecorded spoilage surfaces only later as unexplained monthly shrinkage. An inventory count is mistyped and self-corrects the following night. Each defect is drawn from its own independently keyed stream and simultaneously written to a separate answer key recording exactly what was changed, so the exercise of reconciling the messy records back to the truth is itself gradable against a known answer, and the underlying simulated reality that Section 5’s counterfactual machinery depends on is never disturbed by paperwork noise that exists only in the documents an analyst receives.

This drift has its own exact algebra. Let D_{st} denote a product’s cumulative book-minus-true discrepancy on day t , pushed upward by duplicate invoices and unrecorded spoilage and downward by missing invoices and duplicate receipts. The inventory figure an analyst actually sees is book stock rather than true stock, $B_{st} = S_{st} + D_{st} - \sum_{c \leq t} A_{sc}$, where A_{sc} is the cumulative correction posted at each monthly physical count, which snaps the book back to physical truth before it begins drifting anew. Genuine refunds are kept deliberately separate from this recording layer and modeled instead as a true simulation event: after each receipt is finalized, an independently keyed stream decides whether one of its lines is returned, $\Pr(\text{refund} \mid \text{receipt}) = 0.006$, with the return posted days later as its own till transaction, referencing the original receipt, destroying rather than restocking the returned unit.

The same discipline governs money. Revenue is recognized on a cash basis at the moment of sale, at the price actually paid after any live markdown. A voided mis-ring, in which a cashier corrects an immediate scanning error, nets to zero money and zero goods and is distinguishable from a genuine refund by the absence of a reference to an earlier receipt. Book stock is governed by a strict identity, $\text{book}(t) - \text{book}(t - 1) = \text{deliveries}(t) - \text{sales}(t) - \text{write_offs}(t)$, with any deviation traceable either to a deliberate recording defect or to the monthly physical count. Perishable spoilage and monthly count corrections are kept in strictly separate categories because they measure two different things, genuine physical waste and pure bookkeeping drift, and conflating them would double-count the same euros. The monthly ledger aggregates revenue, procurement, rent, wages, utilities, storage, promotional costs, net tax, and credit interest into a single bottom line, made explicit at the annual horizon as

$$\begin{aligned} \text{ProfitBeforeTax} \stackrel{\Leftarrow}{=} & \sum \text{Revenue} - \sum (\text{Procurement} + \text{Rent} + \text{Wages}) \\ & - \sum (\text{PayrollTax} + \text{Utilities} + \text{Storage}) \\ & - \sum (\text{Flyers} + \text{VAT} + \text{CreditInterest}) - \text{Openings}, \end{aligned}$$

with profit tax levied only on a genuine surplus, $\text{ProfitTax} = 0.20 \times \max(0, \text{ProfitBeforeTax})$ and $\text{ProfitAfterTax} = \text{ProfitBeforeTax} - \text{ProfitTax}$, so a loss year triggers no tax liability but also earns no rebate, mirroring the asymmetric treatment a real small proprietorship faces. The rule that makes every part of this paper trustworthy at once is this: the owner's own monthly ledger is always correct about money, and every other document — particularly the raw till receipts and the raw supplier invoices — is the analyst's own paperwork to reconcile against it.

7 Design Principles and Notation

A few conventions recur so often in the sections above that it is worth naming them once, plainly, rather than re-explaining them at every occurrence.

7.1 The three-question discipline

Every stochastic element in this paper, from a customer's weekly grocery budget to the day a storm arrives, was chosen by answering the same three questions, in the same order. What values can the variable actually take — its *support*? A probability must live on $(0, 1)$; a count must be a nonnegative integer; a cost must be strictly positive. What *generative story* actually produces it? A quantity built from many small effects adding together behaves differently from one built from many small factors multiplying together, and both differ from one built from a fixed number of independent yes-or-no trials. And, once support and story have narrowed the

field, which distribution within it is *maximum entropy* — adds the least unwarranted structure beyond the facts actually being claimed? This discipline explains a pattern that recurs throughout this paper: a budget built from many small multiplicative factors is LogNormal; a probability such as a schedule-adherence rate is Beta; a count of rare independent events such as a shock arrival is Poisson. None of these choices are arbitrary or repeated by habit; each answers the same three questions afresh in its own context.

7.2 Notation

Distributions are written with their natural parameters throughout, so a Normal distribution is always $\text{Normal}(\mu, \sigma)$ and never $\text{Normal}(\mu, \sigma^2)$. The symbol $\sim$ denotes a stochastic draw; $=$ denotes a deterministic definition; and $\stackrel{\leftarrow}{=}$ is reserved for the cost and accounting identities, where the direction of causation is itself the point being made. A hat, $\hat{\cdot}$, always marks a belief held by the owner, as opposed to the corresponding true value he cannot observe. Locations are indexed by l , categories by c (twelve, matching a real product catalog), products by s , customers by i , and days by t ; weeks by w , months by m , and shock events by k . A small number of letters are reused across sections for unrelated quantities exactly as they are reused in the underlying implementation — α , m , and χ each mean something different depending on where they appear — and every such reuse is flagged in prose at the point it recurs.

7.3 Reproducibility

The entire model is driven by a single seeded pseudorandom generator, so the whole three-year history is reproducible exactly from one integer, and the keyed-randomness discipline of Section 5 is what turns that reproducibility into genuine counterfactual power rather than mere determinism.

8 Discussion

Four choices recur across every section of this paper and are worth drawing out explicitly, because they explain why the architecture as a whole looks the way it does rather than some equally plausible alternative.

The first is that every random variable is defended on its own terms by the three-question discipline, rather than by convention or convenience, which means the size and shape of every planted pattern in the data — the degree of overstocking caused by Henrik’s optimism, the seasonal swing in a particular product’s sales — can in principle be computed exactly from the model’s own parameters rather than merely observed empirically after the fact.

The second is the strict separation between the micro layer of Section 3 and the macro layer of Section 4: what is decided fresh each time the world runs, and what is written down in

advance and never listens to anything the simulated people do. This is what allows weather and macroeconomic shocks to function as instruments and natural experiments, and it is what allows an owner's documented mistakes to be genuine, traceable consequences of otherwise reasonable behavior rather than arbitrary flaws inserted for pedagogical convenience.

The third is the strict separation, laid out in Section 6, between the underlying physical reality of the simulated world, always internally consistent and exactly conserved, and the paperwork describing that reality, which is allowed to be imperfect in the same limited, realistic ways real business paperwork is imperfect. This is what allows the counterfactual machinery of Section 5 to remain exact even in the presence of realistic recording noise, because that noise is injected only at the point of export and never touches the simulated state itself.

The fourth is that a small number of specific, deliberate gaps between belief and truth — an owner's demand estimate, his forecasting rule, his tax strategy, his capital-allocation decision — are each introduced once, defended on realistic grounds, and then left undisturbed rather than corrected. This is a deliberate design stance rather than an oversight: the entire point of this construction is to produce a world in which the distance between what an agent believes and what is true can be measured precisely, and that distance can only be measured if the model resists the temptation to make its own protagonist smarter than the premise requires.

9 Conclusion

This paper has told the story of a simulated grocery store as a collection of decisions rather than a chronology of software phases: what a customer wants at a shelf, what an owner believes before a single sale exists, how that owner adjusts once real sales do exist, when he dares to risk capital on growing further, and how a whole neighborhood drifts in and out of being his customers, set against a macro layer that is scripted, deterministic, and never listens to any of it. Every one of those decisions was built the same way: from the plain intuition a real person would give you if you asked them why, to the precise mathematical object that intuition becomes. The result is a world in which an observable pattern — a seasonal swing in ice cream sales, an overstocked category, a capital decision that looks reasonable at the time and questionable in hindsight — always traces back to one of these decisions or scripts rather than to an arbitrary injection of noise, and in which the gap between what somebody believed and what was actually true can, for the first time, be measured rather than merely argued about. This paper's job was only ever to explain why each of those mechanisms is there, and what it is like to live inside the economy they add up to.

References

- Dwork, C. (2006). Differential privacy. In *Proceedings of the 33rd International Colloquium on Automata, Languages and Programming (ICALP)*, Part II, Lecture Notes in Computer Science vol. 4052, 1–12. Springer.
- McFadden, D. (1974). Conditional logit analysis of qualitative choice behavior. In P. Zarembka (Ed.), *Frontiers in Econometrics*, 105–142. Academic Press.
- Orcutt, G. H. (1957). A new type of socio-economic system. *Review of Economics and Statistics*, 39(2), 116–123.
- Patki, N., Wedge, R., and Veeramachaneni, K. (2016). The synthetic data vault. In *Proceedings of the IEEE International Conference on Data Science and Advanced Analytics (DSAA)*, 399–410.
- Sarasvathy, S. D. (2001). Causation and effectuation: Toward a theoretical shift from economic inevitability to entrepreneurial contingency. *Academy of Management Review*, 26(2), 243–263.
- Xu, L., Skoularidou, M., Cuesta-Infante, A., and Veeramachaneni, K. (2019). Modeling tabular data using conditional GAN. In *Advances in Neural Information Processing Systems 32 (NeurIPS 2019)*, 7335–7345.